Measuring Urban Exposure to Environmental Hazards

Understanding where hazards intersect with urban life



Understanding Hazard Exposure

Exposure occurs where hazards overlap with people, places, and assets.

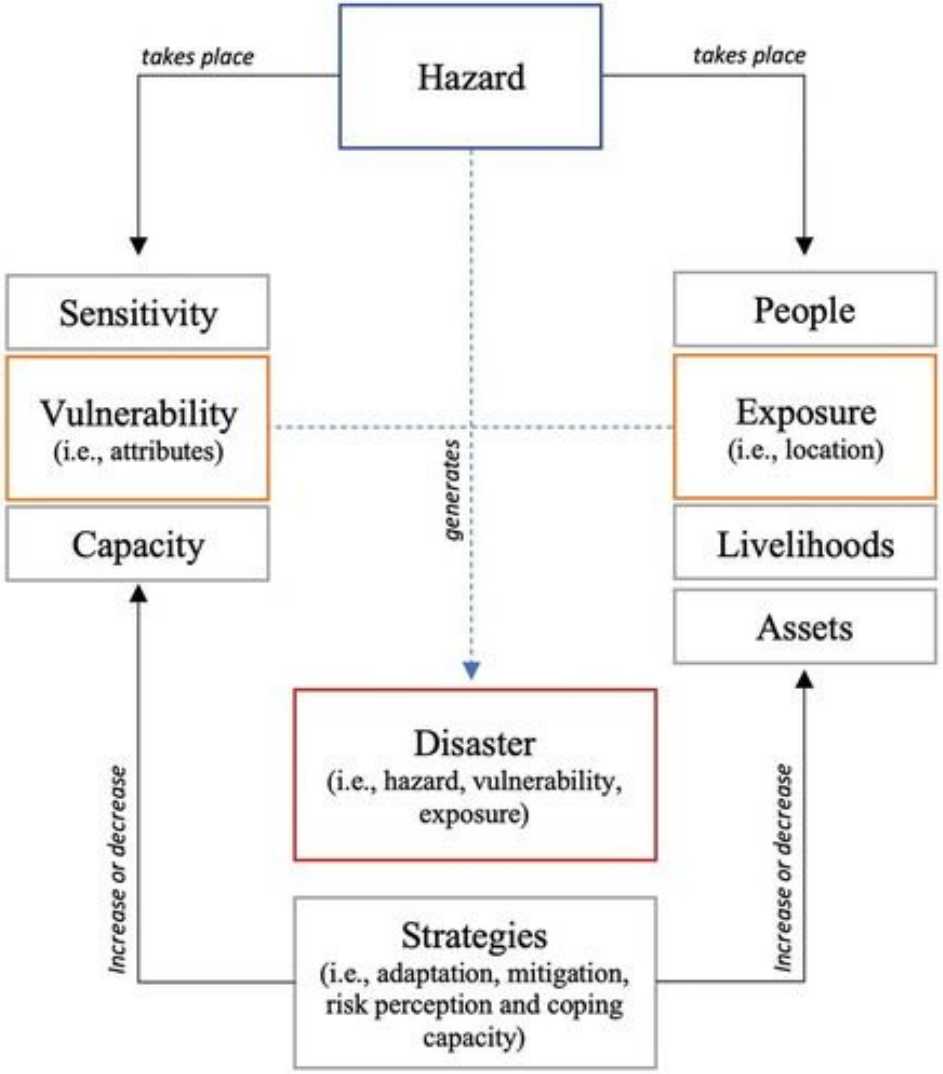


Fig 1. Alves et al. (2021) Conceptualisation of vulnerability and exposure as ‘hazard-specific’ components.

Hazards we will examine…

Three hazards, one question: How can exposure be measured?

Extreme Heat

Wildfires

Flooding

Problem Statement

Environmental hazards do not affect urban areas equally



Hazards differ

Flooding, extreme heat, and wildfire occur at different intensities, times, and scales.



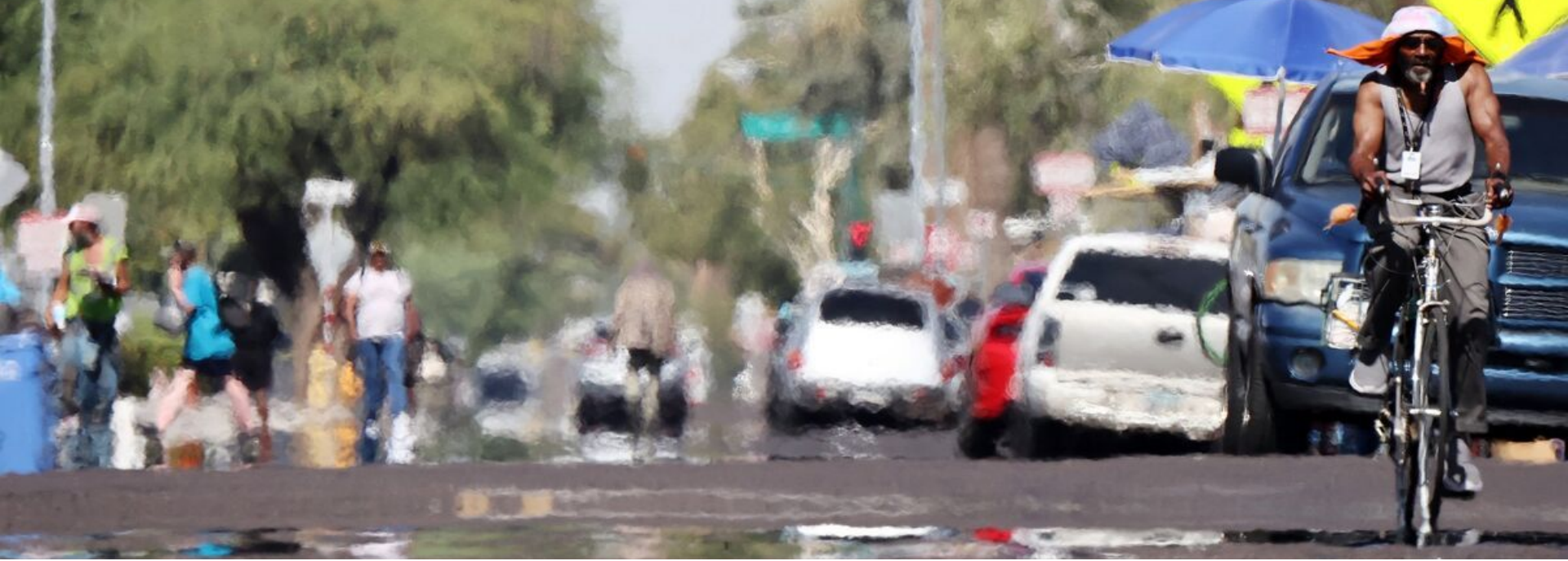
Exposure differs

Where people, buildings, and infrastructure are located determines how exposed these places are to certain hazards.



Impact differs

The built environment and access to critical services can influence how communities experience and respond to hazards.



Research Question 01

How can we measure which populations and urban areas experience the greatest exposure to environmental hazards?

Precedent 01

Using Spatial Data to reveal unequal heat exposure across Atlanta, GA

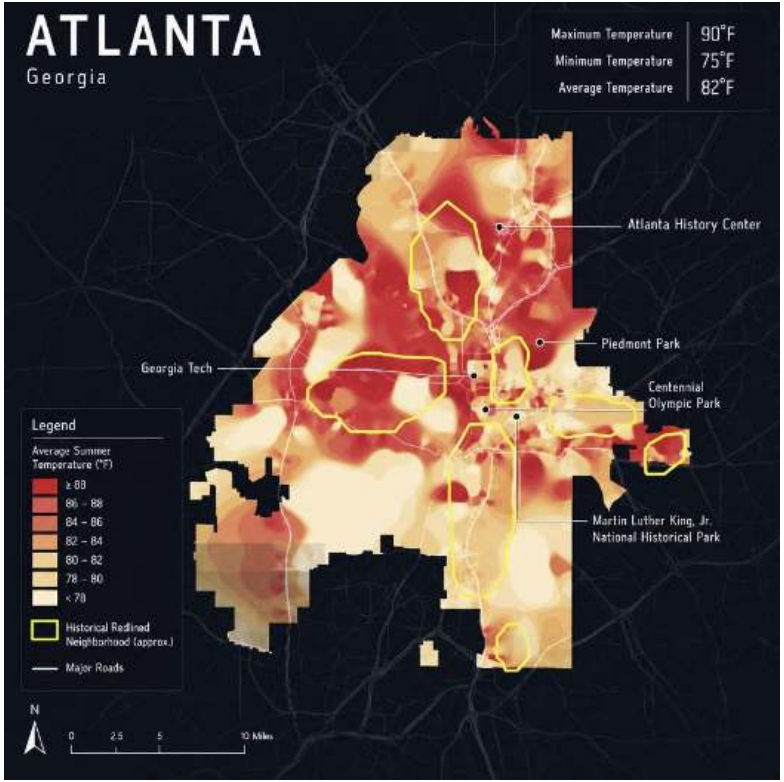
What they measured?

How historic redlining and built-environment conditions influence unequal heat exposure across Atlanta neighborhoods.

How they measured it

- Temperature data
- Tree canopy and green space
- Land cover
- Historic redlining maps

Researchers used GIS to compare neighborhood temperatures with vegetation, land cover, and historic redlining patterns.



Implications

- Historically redlined neighborhoods experience greater heat exposure.
- These areas often have less tree canopy and green space.
- Past planning decisions continue to influence environmental conditions today.
- Unequal access to cooling features increases heat exposure in already vulnerable neighborhoods.

Applying the Measurements to CU Boulder

Temperature / Heat Exposure

- Map surface temperatures across campus
- Identify **hot spots** such as large plazas, parking lots, and highly developed areas
- Compare temperatures between different parts of campus

Tree Canopy + Green Space

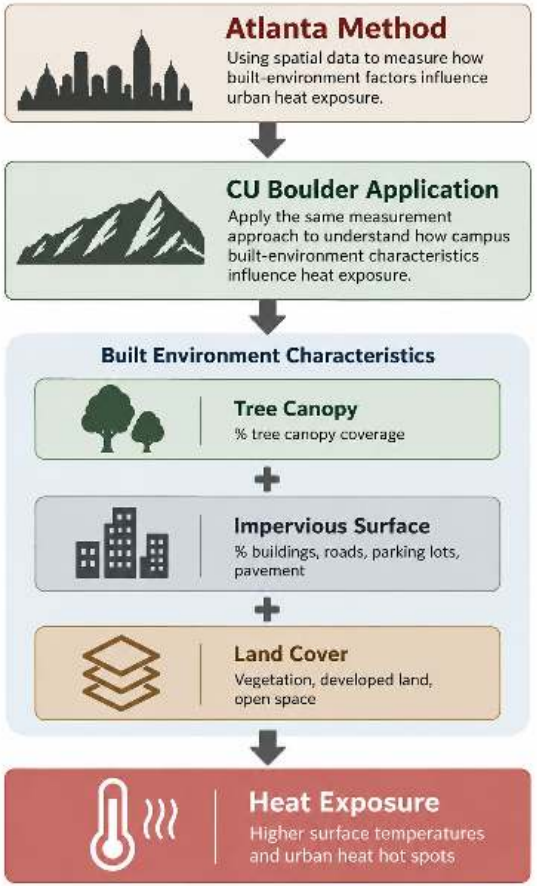
- Calculate **tree canopy coverage (%)** across campus
- Map lawns, fields, landscaped areas, and other vegetation
- Compare areas with high and low canopy to temperature

Impervious Surfaces

- Measure the percentage of **buildings, sidewalks, roads, plazas, and parking lots**
- Identify areas with large concentrations of heat-absorbing surfaces
- Compare impervious surface coverage with heat exposure

Campus Comparison

- Divide CU Boulder into smaller study areas or zones
- Compare **temperature + canopy + impervious surface** between them
- Identify which built-environment characteristics are most associated with hotter areas





Research Question 02

How can characteristics of the built environment be measured to understand their relationship with hazard exposure?

Precedent 02

Built environment characteristics and influence on hazard exposure.

What they measured?

How vegetation, topography, weather, and development patterns influence wildfire exposure in California communities.

How they measured it

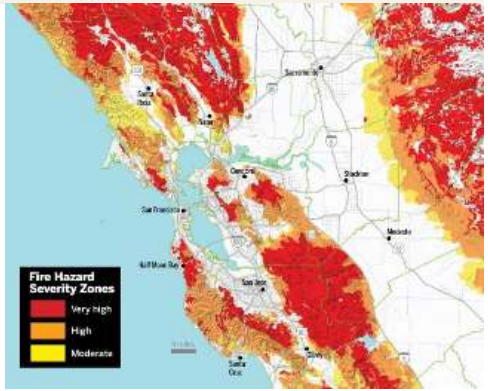
- Fire hazard data
- Vegetation and topography
- Weather conditions
- Development patterns

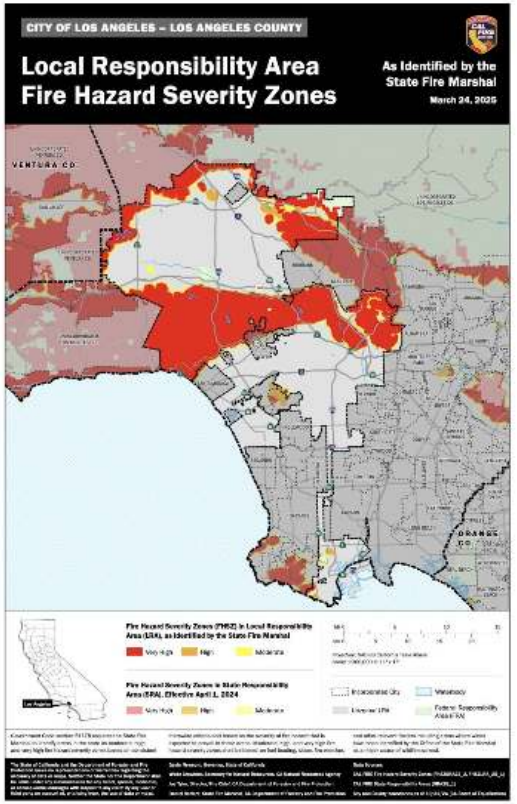
Researchers mapped these factors to classify areas as Moderate, High, or Very High fire hazard.

Implications

Residential development extends into many high-hazard areas. This creates greater wildfire exposure in some desirable and sought-after housing locations







Applying the Measurements to CU Boulder

Vegetation

- Map vegetation type and density around campus
- Identify areas with heavier potential fuel loads
- Compare landscaped areas with surrounding natural vegetation

Topography

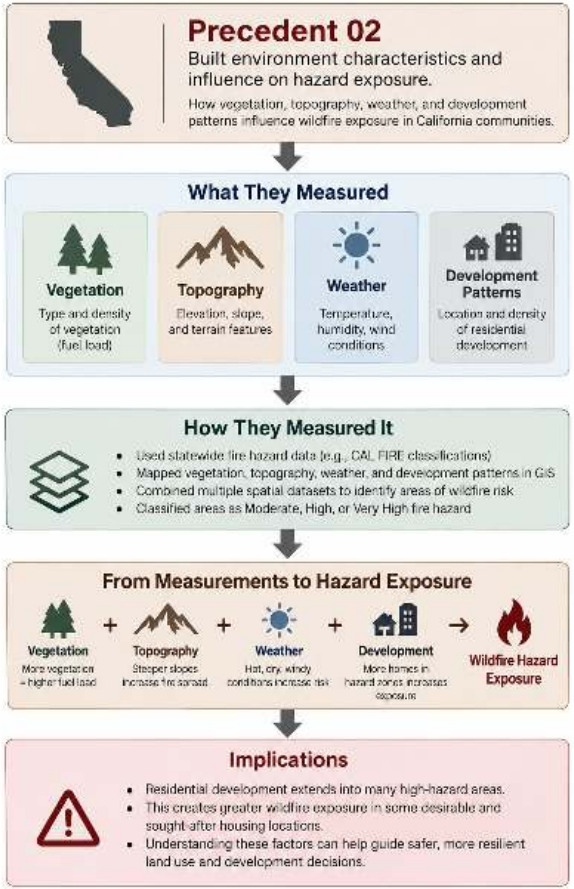
- Measure elevation and slope
- Identify steeper areas where wildfire can spread more rapidly
- Examine how campus development interacts with surrounding terrain

Weather Conditions

- Analyze temperature, humidity, and wind
- Identify conditions associated with increased wildfire risk
- Consider how wind could influence fire movement toward developed areas

Development Patterns

- Map buildings, roads, and developed areas
- Measure proximity between development and wildfire-prone vegetation
- Identify locations where people and structures have greater potential exposure





Research Question 03

How can access and response time to critical services be measured in areas exposed to environmental hazards?

Precedent 03

Evaluating how flood exposure can disrupt access to emergency response services.



2013 Boulder, CO Flood

What they measured?

The percentage of buildings within each census tract that could experience at least 1 foot of flooding from a Hurricane Eta-type of storm.

- Flood model data
- Building footprints

How they measured it

Researchers mapped the percentage of buildings in each census tract that could experience 1 foot of flooding.

Implications

GIS technology would serve to prioritize high valued roads and highways. This could be formalized on their website and should be outlined in their emergency response documentation.

Applying the Measurements to CU Boulder

Vegetation

- Map vegetation type and density around campus.
- Identify areas with dry vegetation and potential fuel loads.
- Locate buildings and pedestrian routes near wildfire-prone vegetation.

Topography

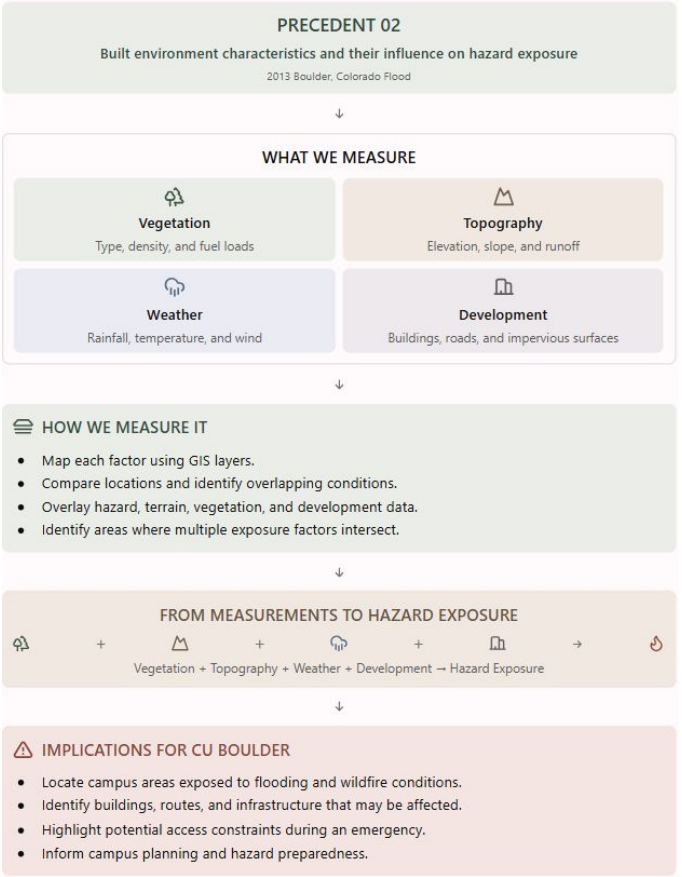
- Map elevation, slope, and low-lying areas.
- Identify steep slopes that can accelerate runoff.
- Examine how campus terrain directs water toward buildings and pathways.

Weather Conditions

- Analyze rainfall intensity, temperature, and wind.
- Identify areas vulnerable to heavy rainfall and flash flooding.
- Consider how wind and dry conditions could increase wildfire exposure.

Development Patterns

- Map buildings, roads, parking lots, and other impervious surfaces.
- Identify infrastructure near creeks, drainage routes, and flood-prone areas.
- Examine how development patterns affect runoff, access, and emergency movement.



Summary of Research

People

Who is exposed?

- Overall population
- Vulnerable groups
- Neighborhood patterns

Place

What makes an area more exposed?

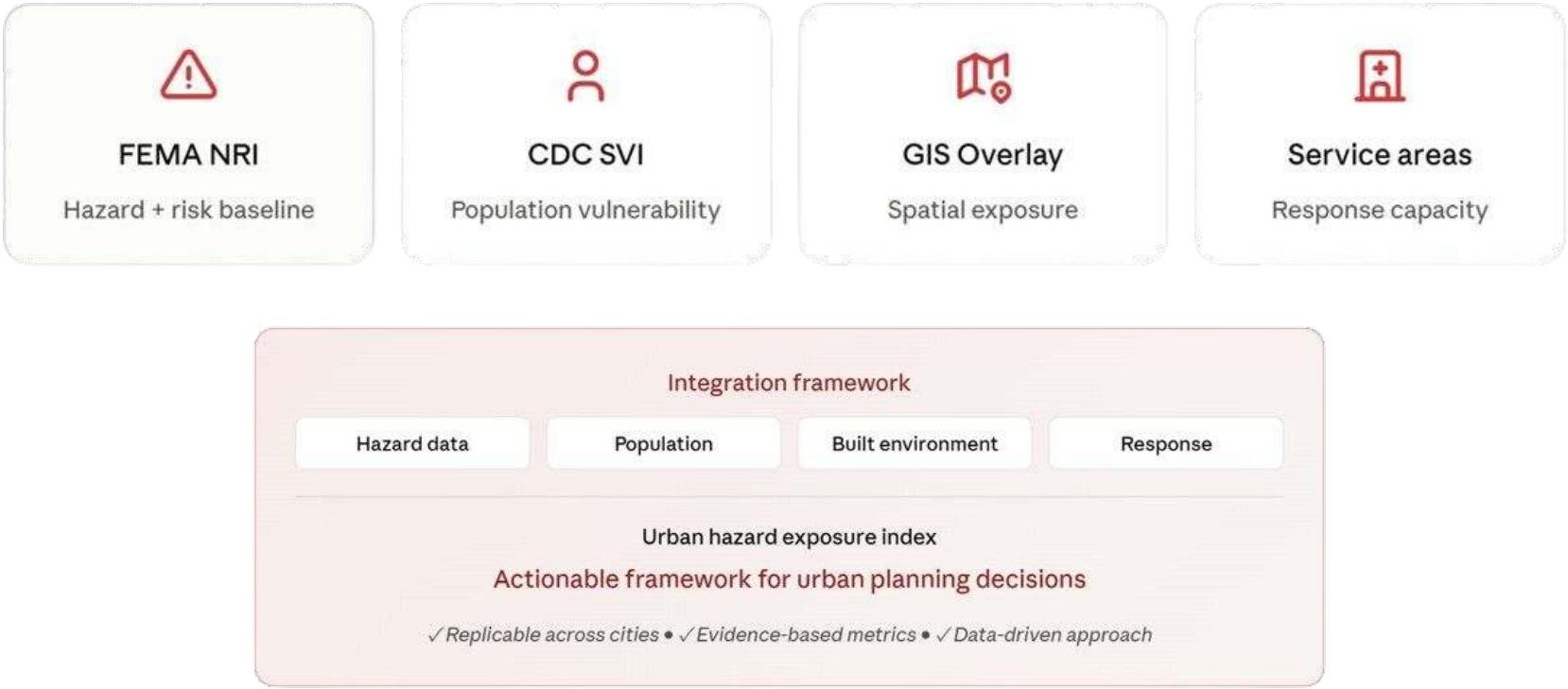
- Buildings
- Infrastructure
- Land use
- Built environment

Response

What happens when a hazard occurs?

- Critical services
- Emergency response
- Transportation access

Assessment Tools



CDC/ATSDR Social Vulnerability Index

FEMA National Risk Index

ArcGIS Service Area Analysis

Thank You